SUPERSTORE SALES FORECASTING

SQL & Power BI Analysis Report

Dataset	Superstore Sales Dataset (2014–2017) (Kaggle)
Tools Used	PostgreSQL | Power BI
Analysis Type	Superstore Sales Forecasting
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1. Project Overview

This report presents the findings from a sales analysis of the Superstore dataset, covering four years of transactional data from 2014 to 2017. The goal of this project was to explore sales performance, identify patterns across product categories and regions, and build a 12-month revenue forecast to support business planning.

The analysis was completed in two stages:

- **Stage 1 — Data Extraction:** SQL queries were written in PostgreSQL to extract and explore the data, covering KPIs, regional breakdowns, category profitability, and monthly trends.
- **Stage 2 — Visualisation:** The SQL results were loaded into Power BI Desktop to build an interactive 3-page dashboard covering Sales Performance, Category Analysis, and a 12-Month Sales Forecast.

2. Dataset Overview

The Superstore dataset is a widely used retail dataset containing order-level sales records for a fictional US-based retail company. It includes information on products, customers, regions, and financials.

Key Fields Used

- order_date, ship_date — used for time-series and monthly trend analysis
- sales, profit, discount, quantity — core financial metrics
- category, sub_category — product hierarchy for category-level analysis
- region, state, city — geographic breakdown
- segment — customer type: Consumer, Corporate, Home Office
- customer_id, order_id — used for counting unique customers and orders

Data Preparation

The raw dataset was imported into PostgreSQL as a CSV file. The order_date and ship_date columns were initially stored as VARCHAR and converted to DATE format using the TO_DATE() function. Basic data validation checks were performed, including reviewing missing values, duplicate records, and date consistency. No major data issues were identified.

3. SQL Analysis

Eight SQL queries were written to extract insights from the dataset. Each query was designed to answer a specific business question and feed directly into the Power BI dashboard.

3.1 Business Overview KPIs

The first query calculated the four-headline metrics used as KPI cards on the dashboard.

Total Revenue	**$2,297,201**	Total Profit	**$286,398**
Total Orders	**5,009**	Total Customers	**793**

Overall profit margin across the full dataset is 12.5%, suggesting there is meaningful variation at the category and product level worth investigating further.

3.2 Revenue and Profit by Category

This query compared the three product categories by revenue and profit to identify where the business earns its money most efficiently.

Category	Total Revenue	Total Profit	Approx. Margin
Technology	$836,154	$145,455	17.4%
Furniture	$741,999	$18,451	2.5%
Office Supplies	$719,047	$122,491	17.0%

Technology and Office Supplies both achieve margins of approximately 17%, while Furniture earns only 2.5% despite being the second-highest revenue category. This gap between revenue and profit in Furniture is a significant finding that was explored further in the sub-category analysis.

3.3 Revenue and Profit by Region

This query identified regional performance to highlight where the business is strongest and where there may be opportunity to improve.

Region	Total Revenue	Total Profit
West	$725,458	$108,419
East	$678,781	$91,523
Central	$501,240	$39,706
South	$391,722	$46,750

The West region leads on both revenue and profit. The Central region generates the second-highest revenue but the lowest profit, suggesting a discount or pricing issue. The South has lower revenue but a relatively better profit ratio than Central.

3.4 Monthly Revenue Trend

Monthly revenue was tracked across all 48 months in the dataset. This data was used as the base for the 12-month forecast on Page 3 of the dashboard. A consistent seasonal pattern was identified, with revenue peaking in Q4 each year — particularly in November and December.

3.5 Revenue and Profit by Sub-Category

Drilling into sub-categories revealed which products were driving and damaging profitability within each category.

Sub-Category	Category	Total Revenue	Total Profit
Copiers	Technology	$149,528	$55,618
Phones	Technology	$330,007	$44,516
Accessories	Technology	$167,380	$41,937
Chairs	Furniture	$328,449	$26,590
Tables	Furniture	$206,966	-$17,726
Bookcases	Furniture	$114,880	-$3,473

Tables is the only sub-category with a significantly negative profit, losing approximately $17,726 across the four-year period. This is likely driven by high discount rates. Copiers, despite lower revenue than Phones, delivers the highest profit of any sub-category.

3.6 Annual Revenue Summary

Year-over-year revenue was calculated to understand the overall growth trend.

Year	Total Revenue	Total Profit	Total Orders
2014	$484,247	$49,544	969
2015	$470,532	$61,619	1,038
2016	$609,206	$81,795	1,315
2017	$733,215	$93,440	1,687

Revenue grew by 51% from 2014 to 2017, increasing from $484K to $733K. Order volume grew from 969 to 1,687, and profit increased steadily each year. The consistent upward trend across all three metrics supports the 12-month forecast.

3.7 Profit Margin by Customer Segment

A final query was added to compare profitability across the three customer segments, which informed the Profit Margin by Segment chart added to Page 2 of the dashboard.

Segment	Total Revenue	Total Profit	Profit Margin
Home Office	$429,653	$60,299	~14.0%
Corporate	$706,146	$91,979	~13.0%
Consumer	$1,161,402	$134,119	~11.5%

Home Office customers generate the highest profit margin despite the lowest revenue. Consumer is the largest segment by revenue but the least margin efficient. This suggests that targeting Home Office and Corporate segments for premium products could improve overall profitability.

4. Power BI Dashboard

The SQL results were imported into Power BI Desktop and used to build a three-page interactive dashboard. Each page focuses on a different aspect of the analysis.

Page 1 — Sales Performance Overview

Page 1 provides a high-level view of business performance across the full 2014–2017 period. It includes five KPI cards, a monthly revenue trend line, a revenue by year bar chart, and regional breakdowns for both revenue and profit.

Key Insight

Revenue grew 51% over four years, from $484K in 2014 to $733K in 2017. Growth accelerated in Q4 each year, suggesting a consistent seasonal demand pattern.

Business Action

Prioritise inventory planning and marketing spend in Q3 to capture peak season demand (Aug–Nov). The West region leads both revenue and profit — it should anchor any regional growth strategy.

Page 2 — Category & Product Analysis

Page 2 examines performance at the category and sub-category level. It includes a Revenue vs Profit by Category chart, a Profit by Sub-Category bar chart, and a Profit Margin by Segment chart. The segment chart replaced an earlier donut chart showing revenue by segment, as it provides more actionable insight.

Key Insight

Furniture generates $742K in revenue but only $18K in profit (2.4% margin). Tables alone lose ~$17K, making it the single most profit-damaging sub-category.

Business Action

A review of Table pricing and discount rates would be a logical next step. Redirect promotional spend toward Copiers and Accessories — the highest-profit sub-categories — to improve overall margin.

Page 3 — Sales Forecast (12-Month Outlook)

Page 3 uses exponential smoothing forecasting to project monthly revenue 12 months beyond December 2017. The seasonal cycle was manually set to 12 points rather than auto-detect, based on

the clearly visible Q4 revenue peak repeating across all four years of historical data. A 95% confidence interval was applied and the uncertainty band deliberately enabled, to communicate the honest range of outcomes rather than imply false precision. The model projects revenue exceeding $100K/month by Q4 2018, consistent with four consecutive years of approximately 20% annual growth.

Key Insight

Revenue is forecast to exceed $100K/month by Q4 2018, extending four consecutive years of ~20% annual growth. Seasonal peaks suggest Q4 will again be the highest-demand period.

Business Action

Begin capacity and staffing planning now to meet projected Q4 2018 demand. Investment in Technology products and the West region offers the strongest return, based on consistent margin and volume performance.

5. Key Findings Summary

The following findings emerged from the combined SQL and Power BI analysis:

- **Growth trend:** Revenue grew 51% from 2014 to 2017, driven by consistent increases in order volume and average order value.
- **Seasonality:** Q4 is consistently the strongest sales period each year, with November and December accounting for a disproportionate share of annual revenue.
- **Regional performance:** The West region is the top performer on both revenue ($725K) and profit ($108K). Central underperforms on profit relative to its revenue.
- **Category efficiency:** Technology and Office Supplies both achieve ~17% margins. Furniture earns only 2.5% despite being the second-largest revenue category.
- **Sub-category issue:** Tables are the single most profit-damaging sub-category, losing $17,726 over four years. Excessive discounting is the most likely cause.
- **Segment insight:** Home Office customers are the most margin-efficient segment at ~14%, despite generating the least revenue.
- **Forecast:** Based on the historical growth trend, revenue is forecast to exceed $100K/month by Q4 2018.

6. Limitations & Next Steps

Limitations

- The dataset covers only 2014–2017, so the forecast is based on a relatively short historical window. External factors such as market conditions or competitor activity are not accounted for.
- Discount data is available at the order line level, but a full discount impact analysis was outside the scope of this project. A deeper dive into the relationship between discount rate and profit margin could add further value.

- The dataset is fictional and based on a US retail context. Findings are directional rather than actionable for a real business.

Suggested Next Steps

- Conduct a discount rate analysis to quantify how much revenue is being given away on Tables and other low-margin sub-categories.
- Segment the West region further to understand which states and cities are driving its strong performance.
- Extend the forecast model with external seasonality inputs to improve confidence interval accuracy.

7. Tools & Skills Demonstrated

Area	Detail
SQL (PostgreSQL)	Table creation, date type conversion, aggregations, GROUP BY, ORDER BY, DATE_TRUNC, EXTRACT, ROUND
Power BI	Data modelling, DAX measures, KPI cards, bar/line charts, donut chart, forecast visual, slicers, multi-page layout
Data Analysis	KPI calculation, trend analysis, category profitability, regional comparison, customer segmentation, forecasting
Communication	Insight and action box copy, structured report writing, dashboard storytelling